

Gone Before the Truck

Auditing How Much of a Council's Hard-Rubbish Diversion Figure Was Already Taken by Hand

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The problem

Councils report a hard-rubbish diversion rate as though every item logged “not landfilled” passed through the contracted service. Little is known about how much of that figure is already gone from the verge by hand before the truck turns into the street.

Research questions

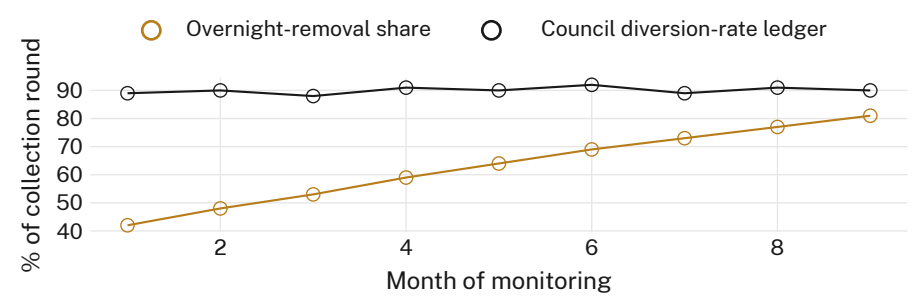
- What share of verge-side items disappear before the truck arrives?
- How does that share compare with the council's own round-level diversion-rate entries?
- Does the ledger's diversion figure distinguish contracted processing from pre-truck removal?

Study design

- n = 214 verge piles across 11 suburbs · item-level photo audit at set-out, dawn, and truck arrival · 9-month monitoring window
- Each item's disappearance timestamp matched against the council's own diversion-rate ledger entry for that collection round
- Collection roster, truck schedule, and ledger formula held constant; no round re-timed or re-audited for the study

What we found

Results suggest a growing majority of the logged diversion figure is attributable to overnight removal rather than the contracted collection. Agreement between the ledger's stated diversion rate and the round's audited contracted-processing share falls across the monitoring window.



Overnight-removal's share of the logged diversion rate rose from 42% in month 1 to 81% by month 9 (n = 214 piles across 11 suburbs, monitored February–October 2025), while the council's own diversion-rate ledger entry moved by less than three percentage points across the same period.

Implications

A diversion rate that holds steady while its underlying cause quietly changes hands is measuring the wrong thing precisely. The ledger cannot currently tell a contractor's load from a stranger's trailer, and the round-level formula has no field for who actually did the diverting. Next: test whether the same drift holds for the council's e-waste and green-waste collection figures.

Selected references

1. Wilson, D. C., Velis, C., & Cheeseman, C. (2006). Role of informal sector recycling in waste management in developing countries. *Habitat International*, 30(4), 797–808. doi:10.1016/j.habitatint.2005.09.005
2. Velis, C. A. (2017). Waste pickers in Global South: informal recycling sector in a circular economy era. *Waste Management & Research*, 35(4), 329–331. doi:10.1177/0734242X17702024
3. Wibin, P., & Kalina, M. (2025). Informal waste pickers and non-motorised transport: a mini-review of contemporary literature. *Waste Management & Research*. doi:10.1177/0734242X251361730
4. Cook, E., Silva de Souza Lima Cano, N., & Velis, C. A. (2024). Informal recycling sector contribution to plastic pollution mitigation: a systematic scoping review and quantitative analysis of prevalence and productivity. *Resources, Conservation and Recycling*, 206, 107588. doi:10.1016/j.resconrec.2024.107588
5. Zhang, S., & Ma, J. (2026). Urban waste management inequities in New York city: a case study on spatial patterns of controlled and uncontrolled street waste and their associated factors. *Waste Management*, 212, 115333. doi:10.1016/j.wasman.2026.115333

We thank the residents of the eleven monitored streets; item photographs de-identified at capture, no collector identity retained.

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